

Equi Join in DBMS

Comprehensive, High-Yield Exam & GATE Preparation Notes

1. WHAT IS EQUI JOIN?

An **Equi Join** is a type of join in which the joining condition uses exclusively the **equality operator (=)**.

★ Exam-Ready Definition

An Equi Join is a join in which two relations are joined using an equality condition between attributes. The equality condition can be established between attributes of the same domain/meaning, and technically it does not have to be restricted to common-named attributes.

2. EXAMPLE DATASET

Consider the following two relational tables:

Employee Table

Emp_No	Emp_Name	Address
1	Ram	Delhi
2	Varun	Chandigarh
3	Ravi	Delhi
4	Amrit	Delhi

Department Table

Dept_No	Location	Emp_No
D1	Delhi	1
D2	Pune	2
D3	Patna	4

Note: *Department.Emp_No* refers to the employee associated with that department.

3. PROBLEM STATEMENT & CONDITIONS

Problem: Find the employee name whose department location matches the employee's residential address.

To produce correct results, we must satisfy **two joint conditions**:

- **Condition 1 (Association):** Employee must be assigned to that department:

Employee.Emp_No = Department.Emp_No

- **Condition 2 (Location Match):** Employee's address must equal the department's location:

Employee.Address = Department.Location

4. SQL QUERY

Using standard relational join conditions (Both conditions use the = operator):

```
SELECT Employee.Emp_Name
FROM Employee, Department
WHERE Employee.Emp_No = Department.Emp_No
AND Employee.Address = Department.Location;
```

5. STEP-BY-STEP QUERY EXECUTION

1. **SELECT:** Output column required is *Employee.Emp_Name*.
2. **FROM:** Combines *Employee* and *Department*. Conceptually creates Cartesian Product (*Employee* × *Department*).
Total combinations evaluated = $4 \times 3 = 12$ rows.
3. **WHERE Conditions:** Filters rows satisfying (*Condition 1*) AND (*Condition 2*).

6. ROW EVALUATION & TRACE

Employee 1: Ram (Emp_No = 1, Address = Delhi)

✓ SELECTED

Matches Dept D1 (Emp_No = 1, Location = Delhi)

- Cond 1: $1 = 1$ (TRUE) | Cond 2: *Delhi* = *Delhi* (TRUE)
- Result: **TRUE AND TRUE = TRUE**

Employee 2: Varun (Emp_No = 2, Address = Chandigarh)

✗ REJECTED

Matches Dept D2 (Emp_No = 2, Location = Pune)

- Cond 1: $2 = 2$ (TRUE) | Cond 2: *Chandigarh* = *Pune* (FALSE)
- Result: **TRUE AND FALSE = FALSE**

Employee 3: Ravi (Emp_No = 3, Address = Delhi)

✗ REJECTED

No row in Department with *Emp_No* = 3.

- Result: **FALSE (No matching association)**

Employee 4: Amrit (Emp_No = 4, Address = Delhi)

✗ REJECTED

Matches Dept D3 (Emp_No = 4, Location = Patna)

- Cond 1: $4 = 4$ (TRUE) | Cond 2: *Delhi* = *Patna* (FALSE)
- Result: **TRUE AND FALSE = FALSE**

7. FINAL OUTPUT RESULT

Only **Ram** satisfies both conditions simultaneously:

Emp_Name

Ram

8. EQUI JOIN VS. NATURAL JOIN (GATE SPECIAL)

Feature	Equi Join	Natural Join
Operator	Explicit = operator	Equality used implicitly by DBMS
Explicit Condition?	✓ Yes (ON clause or WHERE clause)	✗ No (Automatic based on schema)
Attribute Selection	Explicitly specified attributes	All same-named common attributes
Same Column Names?	✗ Not required (e.g. <i>E.ID</i> = <i>D.Emp_ID</i>)	✓ Required for automatic matching
Duplicate Columns	✗ Retained in output (<i>SELECT *</i>)	✓ Duplicate join columns auto-removed
SQL Example	<i>SELECT * FROM E JOIN D ON E.id = D.id</i>	<i>SELECT * FROM E NATURAL JOIN D</i>

9. MODERN SQL SYNTAX & IMPORTANT EXAM TRAP

Modern ANSI SQL Syntax

```
SELECT E.Emp_Name
FROM Employee E
JOIN Department D
  ON E.Emp_No = D.Emp_No
 AND E.Address = D.Location;
```

⚠ Important Exam Trap:

The statement "In Equi Join, equality must always be applied to attributes with identical names" is **FALSE**.

For instance, **ON E.Emp_No = D.Employee_ID** is a perfectly valid Equi Join. What defines an Equi Join is strictly the use of the **equality operator (=)**, regardless of attribute naming.

10. CLASSIFICATION & QUICK MEMORY RULES

An Equi Join is a specific subtype of a **Theta Join (Conditional Join)**:

Equi Join $\subseteq$ Theta Join

JOIN → Combine Tables → Specify Condition → If Condition Uses "=" → EQUI JOIN

🧠 Memory Trick & Definition Summary:

- **One-Line Definition:** An Equi Join is a conditional join that uses the equality operator (=) to match attributes between relations.
- **Golden Rule: EQUI = EQUAL (=).** Whenever join conditions rely exclusively on equality comparisons, it is an Equi Join.